

Comprehensive Notes: Layer Normalization in Transformers

CampusX Lecture Summary

1 Core Prerequisites & Concepts

Normalization in deep learning refers to scaling the inputs or intermediate hidden layer activations to achieve specific statistical properties—typically forcing a mean (μ) of 0 and a standard deviation (σ) of 1.

Why Normalize?

- **Improved Training Stability:** Prevents activations from becoming extremely large or small, mitigating vanishing/exploding gradients.
 - **Faster Convergence:** Smooths out the loss landscape, letting the network utilize higher learning rates and converge in fewer epochs.
 - **Mitigating Internal Covariate Shift:** This phenomenon occurs when the distribution of hidden layer activations constantly shifts during training because preceding weights are being updated. Normalization fixes this baseline distribution.
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2 Batch Normalization (BN) Deep Dive

Batch Normalization operates by standardizing activations **across the batch dimension** for each individual feature/neuron independently.

The Mathematical Formula

For a given mini-batch of size B , and a specific feature activation:

1. **Compute Batch Mean:**

$$\mu_B = \frac{1}{B} \sum_{i=1}^B x_i \quad (1)$$

2. **Compute Batch Variance:**

$$\sigma_B^2 = \frac{1}{B} \sum_{i=1}^B (x_i - \mu_B)^2 \quad (2)$$

3. **Normalize:**

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \quad (3)$$

(where ϵ is a tiny constant added for numerical stability to avoid division by zero).

4. **Scale and Shift:**

$$y_i = \gamma \hat{x}_i + \beta \quad (4)$$

γ (gamma) and β (beta) are **learnable parameters** optimized via backpropagation. They allow the network to undo the normalization if a different distribution is more optimal for performance.

Why Batch Norm Fails in Transformers (NLP)

- **Variable Sequence Lengths:** Sentences in a single batch naturally differ in length. Shorter sequences require heavy padding tokens (zeros). Computing mean and variance across the batch dimension forces padding values into the calculation, distorting the statistical realities of the real words.
 - **Batch Size Dependency:** BN relies on a reasonably large batch size to calculate stable statistics. In large Transformer models with massive parameter footprints, memory constraints often restrict batch sizes to small values, making BN calculations highly volatile.
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3 Layer Normalization (LN) to the Rescue

Layer Normalization solves the flaws of BN by calculating statistics **across the features of a single training example**, completely independent of other items in the batch.

The Mathematical Formula

Instead of averaging across multiple rows (samples), LN calculates the mean and variance across all features (channels/hidden dimensions) for **one single sequence token**.

For a specific sequence token vector x containing D dimensions:

1. **Compute Layer Mean:**

$$\mu_L = \frac{1}{D} \sum_{j=1}^D x_j \quad (5)$$

2. **Compute Layer Variance:**

$$\sigma_L^2 = \frac{1}{D} \sum_{j=1}^D (x_j - \mu_L)^2 \quad (6)$$

3. **Normalize:**

$$\hat{x}_j = \frac{x_j - \mu_L}{\sqrt{\sigma_L^2 + \epsilon}} \quad (7)$$

4. **Scale and Shift:**

$$y_j = \gamma_j \hat{x}_j + \beta_j \quad (8)$$

Just like BN, γ and β are learnable parameters applied element-wise, but they map to the feature dimensions rather than data points.

4 Architectural Comparison: BN vs. LN

5 Summary Placement in Transformers

In the Transformer architecture (both Encoder and Decoder blocks), Layer Normalization is placed directly within the "Add & Norm" sections.

It receives the output of either the Multi-Head Self-Attention block or the Feed-Forward Network, adds it to the original block input via a **Residual Connection** ($Z + X$), and applies Layer Normalization to yield a stable, bounded feature space for subsequent blocks.

Table 1: Comparison between Batch Normalization and Layer Normalization

Metric / Feature		Batch Normalization (BN)	Layer Normalization (LN)
Averaging Direction		Vertically (Across the mini-batch dimension).	Horizontally (Across the features of a single sample).
Dependency		Heavily dependent on the total batch size.	Entirely independent of batch size.
Handling Padding		Vulnerable; padding tokens heavily skew batch stats.	Immune; ignores other sequences' padding.
Primary Domain		Computer Vision (CNNs).	Natural Language Processing (Transformers / RNNs).